Assignment 4

Problem

According to one version of Moore's law, the number of transistors on a state-of-the-art computer microprocessor roughly doubles every two years:

$$C \approx \gamma 2^{A/2}$$

where

- A: the year number in which a microprocessor is introduced
- C: the transistor count of the microprocessor
- γ: a positive constant

Consider the **natural logarithm** of the transistor count: $\log C$

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a. Show that $\log C$ is roughly a simple linear regression and derive the coefficient of A

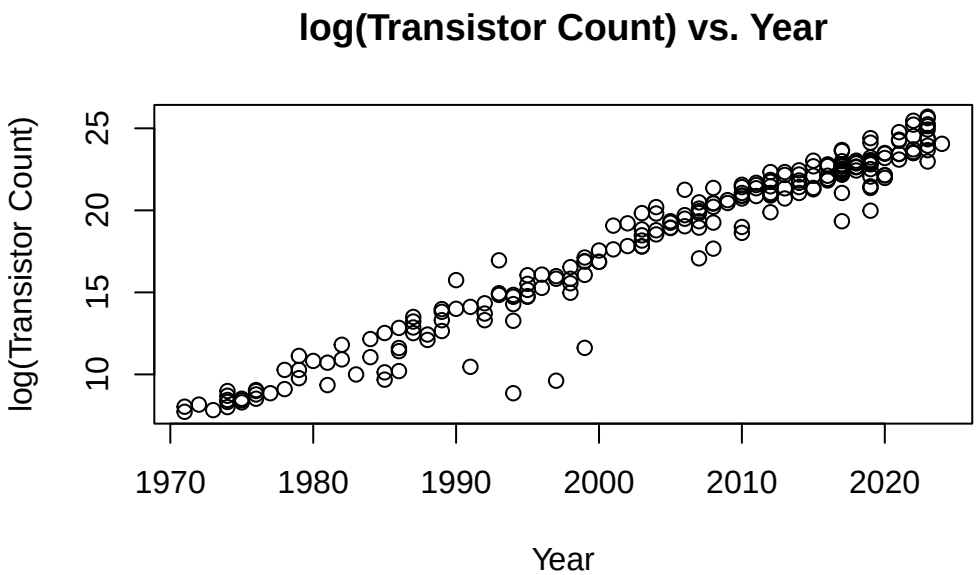
$$\begin{aligned} \log C &\approx \log \gamma + \frac{A}{2} \log 2 \\ &\approx \log \gamma + A\frac{1}{2} \log 2 \end{aligned}$$

Given that the form of a simple linear regression is $y = \beta_1 + x\beta_2$ where β_1 and β_2 are the parameters, $\log C$ is roughly a simple linear regression with:

- $y = \log C$
- $x = A$
- $\beta_1 = \log \gamma$
- $\beta_2 = \frac{1}{2} \log 2 = 0.3465736$ **(the coefficient of A)**

b. Plot the data points as log transistor count versus year

```r
transistor <- read.csv("./Moores Law Data.csv", header = TRUE)
transistor$logC <- log(transistor$TransistorCount)
plot(
    x = transistor$Year,
    y = transistor$logC,
    main = "log(Transistor Count) vs. Year",
    ylab = "log(Transistor Count)",
    xlab = "Year"
)
```



c. Consider a normal-theory simple linear regression model

$$\log C_i \mid \beta, \sigma^2 A_i \sim indep.\ \ N(\beta_1 + \beta_2(A_i - \bar{A}), \sigma^2) \quad i = 1, \ldots, 235$$

where

- $\beta_1, \beta_2 \sim \quad iid.\ N(0, 1000^2)$
- $\sigma^2 \sim \text{Inv-gamma}(0.001, 0.001)$

i. List an appropriate JAGS model

```
model {
    for (i in 1:length(Year)) {
        logC[i] ~ dnorm(beta_1 + beta_2 * (Year[i] - AvgYear[i]), tau)
    }
    beta_1 ~ dnorm(0, 1/1000000)
    beta_2 ~ dnorm(0, 1/1000000)
    tau ~ dgamma(0.001, 0.001)
    sigma.2 <- 1 / tau
}
```

Now, run the model with overdispersed starting points, check convergence, and monitor β_1, β_2, and σ^2 for at least 2000 iterations after burn-in.

```
transistor$AvgYear = mean(transistor$Year)
init.vals <- list(
    list(beta_1=10000, beta_2=10000, tau=0.00001),
    list(beta_1=-10000, beta_2=-10000, tau=0.00001),
    list(beta_1=10000, beta_2=10000, tau=0.01),
    list(beta_1=-10000, beta_2=-10000, tau=0.01)
)

library(rjags)
```

```
Loading required package: coda

Linked to JAGS 4.3.2

Loaded modules: basemod,bugs
```

```
ml <- jags.model("./regression.bug", transistor, init.vals, n.chains=4, n.adapt=1000)
```

```
Warning in jags.model("./regression.bug", transistor, init.vals, n.chains = 4,
: Unused variable "Processor" in data

Warning in jags.model("./regression.bug", transistor, init.vals, n.chains = 4,
: Unused variable "TransistorCount" in data
```

```
Compiling model graph
   Resolving undeclared variables
   Allocating nodes
Graph information:
   Observed stochastic nodes: 235
   Unobserved stochastic nodes: 3
   Total graph size: 876

Initializing model
```

```r
update(ml, 2500)
x <- coda.samples(ml, c("beta_1", "beta_2", "sigma.2"), n.iter=5000)

# verify convergence
gelman.diag(x, autoburnin=FALSE)
```

```
Potential scale reduction factors:

        Point est. Upper C.I.
beta_1           1          1
beta_2           1          1
sigma.2          1          1

Multivariate psrf

1
```

- As shown with the Gelman-Rubin convergence diagnostic, PSRF values equal to 1 for all parameters, indicating that MCMC chains have converged.

ii. List summary for β_1, β_2, and σ^2

```r
summary(x)
```

```
Iterations = 2501:7500
Thinning interval = 1
Number of chains = 4
Sample size per chain = 5000

1. Empirical mean and standard deviation for each variable,
```

```
   plus standard error of the mean:

           Mean       SD  Naive SE Time-series SE
beta_1  18.3956 0.072727 5.143e-04      5.143e-04
beta_2   0.3328 0.004902 3.466e-05      3.467e-05
sigma.2  1.2465 0.116137 8.212e-04      8.212e-04

2. Quantiles for each variable:

           2.5%     25%     50%    75%   97.5%
beta_1  18.2520 18.3473 18.3958 18.445 18.5375
beta_2   0.3231  0.3295  0.3328  0.336  0.3424
sigma.2  1.0376  1.1668  1.2398  1.320  1.4928
```

---

iii. Give the approximate posterior mean and 95% central posterior interval for the slope

- Slope is the statistics related to β_2

Slope Mean

```r
summary(x)$statistics["beta_2", "Mean"]
```

```
[1] 0.3327589
```

Slope 95% Central Posterior Interval

```r
x_mat = as.matrix(x)
quantile(x_mat[,"beta_2"], c(0.025, 0.975))
```

```
     2.5%     97.5%
0.3231325 0.3424146
```

- No, the coefficient of A does not fall within the interval. It goes slightly beyond the upper end, which suggests that the estimated growth in transistor count is slightly lower than Moore's law.

---

iv. Give the approximate posterior mean and 95% central posterior interval for the error variance

Slope Mean

```
summary(x)$statistics["sigma.2", "Mean"]
```

```
[1] 1.246509
```

Slope 95% Central Posterior Interval

```
quantile(x_mat[,"sigma.2"], c(0.025, 0.975))
```

```
    2.5%    97.5%
1.037612 1.492795
```

d. Predict the transistor count

i. Update JAGS model to answer the subsequent questions

```
model {
    for (i in 1:length(Year)) {
        logC[i] ~ dnorm(beta_1 + beta_2 * (Year[i] - AvgYear[i]), tau)
    }
    beta_1 ~ dnorm(0, 1/1000000)
    beta_2 ~ dnorm(0, 1/1000000)
    tau ~ dgamma(0.001, 0.001)
    sigma.2 <- 1 / tau

    mu_2026 <- beta_1 + beta_2 * (2026 - AvgYear[1])
    logC_2026 ~ dnorm(mu_2026, tau)
    C_2026 <- exp(logC_2026)
    C_2026_billions <- C_2026 / 1000000000

    invention_year <- AvgYear[1] - beta_1 / beta_2
}
```

```
ml <- jags.model("./regression_d.bug", transistor, init.vals, n.chains=4, n.adapt=1000)
```

```
Warning in jags.model("./regression_d.bug", transistor, init.vals, n.chains =
4, : Unused variable "Processor" in data

Warning in jags.model("./regression_d.bug", transistor, init.vals, n.chains =
4, : Unused variable "TransistorCount" in data

Compiling model graph
   Resolving undeclared variables
   Allocating nodes
Graph information:
   Observed stochastic nodes: 235
   Unobserved stochastic nodes: 4
   Total graph size: 886

Initializing model
```

```r
update(ml, 2500)
x <- coda.samples(ml, c("C_2026_billions", "invention_year"), n.iter=5000)

# verify convergence
gelman.diag(x, autoburnin=FALSE)
```

```
Potential scale reduction factors:

                Point est. Upper C.I.
C_2026_billions          1          1
invention_year           1          1

Multivariate psrf

1
```

---

ii. List summary

```r
summary(x)
```

```
Iterations = 2501:7500
Thinning interval = 1
Number of chains = 4
Sample size per chain = 5000

1. Empirical mean and standard deviation for each variable,
   plus standard error of the mean:

                   Mean       SD Naive SE Time-series SE
C_2026_billions  231.7 375.3269 2.653962       2.754942
invention_year  1949.3   0.8492 0.006005       0.006005

2. Quantiles for each variable:

                   2.5%     25%  50%    75% 97.5%
C_2026_billions    13.6   56.28  120  259.7  1135
invention_year   1947.6 1948.69 1949 1949.8  1951
```

---

iii. Give an approximate 95% central posterior predictive interval for the transistor count

```r
x_mat <- as.matrix(x)
quantile(x_mat[,"C_2026_billions"], c(0.025, 0.975))
```

```
     2.5%     97.5%
  13.6047 1134.6378
```

---

iv. Explain the following formula and give the 95% central posterior interval

$$\bar{A} - \frac{\beta_1}{\beta_2}$$

Proof

- When the transistor was invented, $C = 1$, which leads to $\log C = 0$
- Therefore,

$$
\begin{aligned}
&\beta_1 + \beta_2 * (A - \bar{A}) = 0 \\
&\beta_2 A = \beta_2 \bar{A} - \beta_1 \\
&A = \bar{A} - \frac{\beta_1}{\beta_2}
\end{aligned}
$$

95% central posterior interval

```
quantile(x_mat[,"invention_year"], c(0.025, 0.975))
```

```
    2.5%    97.5%
1947.558 1950.891
```

Actual year

- Transistor was invented in 1947, which falls within the 95% central posterior interval

e. Check for outliers

$$T(y, X, \theta) = \max_i |\epsilon_i / \sigma|$$

```
ml <- jags.model("./regression.bug", transistor, init.vals, n.chains=4, n.adapt=1000)
```

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Warning in jags.model("./regression.bug", transistor, init.vals, n.chains = 4,
: Unused variable "Processor" in data
```

```
Warning in jags.model("./regression.bug", transistor, init.vals, n.chains = 4,
: Unused variable "TransistorCount" in data
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```
Compiling model graph
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Graph information:
   Observed stochastic nodes: 235
   Unobserved stochastic nodes: 3
   Total graph size: 876

Initializing model
```

```r
update(ml, 2500)
x <- coda.samples(ml, c("beta_1", "beta_2", "sigma.2"), n.iter=5000)
x_mat <- as.matrix(x)
```

i. Compute simulated standardized error vectors ϵ/σ

```r
n <- nrow(x_mat)
K <- length(transistor$logC)

eps_std <- matrix(NA, nrow = n, ncol = K)
for (s in 1:n) {
    beta1 <- x_mat[s, "beta_1"]
    beta2 <- x_mat[s, "beta_2"]
    sigma <- sqrt(x_mat[s, "sigma.2"])

    mu <- beta1 + beta2 * (transistor$Year - transistor$AvgYear)
    eps <- transistor$logC - mu
    eps_std[s,] <- eps / sigma
}
```

---

ii. Compute simulated replicate standardized errors ϵ^{rep}/σ

```r
eps_rep_std <- matrix(NA, nrow = n, ncol = K)
for (s in 1:n) {
    beta1 <- x_mat[s, "beta_1"]
    beta2 <- x_mat[s, "beta_2"]
    sigma <- sqrt(x_mat[s, "sigma.2"])
```

```r
    mu <- beta1 + beta2 * (transistor$Year - transistor$AvgYear)
    y_rep <- rnorm(K, mean = mu, sd = sigma)
    eps_rep_std[s,] <- (y_rep - mu) / sigma
}
```

---

iii. Compute $T(y, X, \theta)$ and $T(y^{rep}, X, \theta)$

$T(y, X, \theta)$

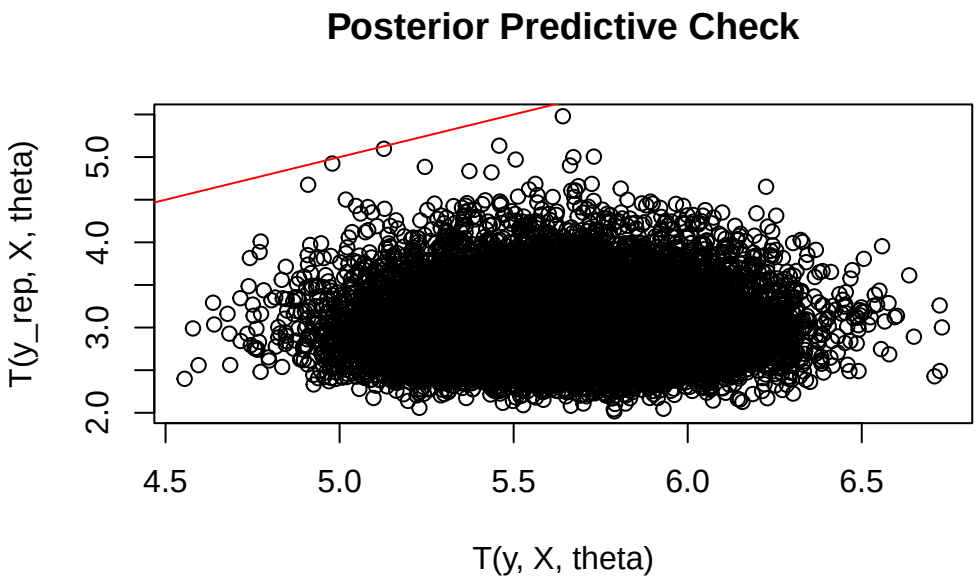
```r
T <- apply(abs(eps_std), 1, max)
```

$T(y^{rep}, X, \theta)$

```r
T_rep <- apply(abs(eps_rep_std), 1, max)
```

iv. Plot T^{rep} vs T

```r
plot(
    T,
    T_rep,
    xlab = "T(y, X, theta)",
    ylab = "T(y_rep, X, theta)",
    main = "Posterior Predictive Check"
)
abline(0, 1, col = "red")
```



Calculate the p-value

```
mean(T_rep >= T)
```

```
[1] 0
```

- As the p-value equals 0, the observed extremes are always greater than or equal to the posterior model extremes. Therefore, we can infer that there must be an outlier in the observed data.

Name the microprocessor that appears to be the most extreme outlier

```
x_stats = summary(x)$statistics
beta1_hat <- x_stats["beta_1", "Mean"]
beta2_hat <- x_stats["beta_2", "Mean"]
sigma_hat <- sqrt(x_stats["sigma.2", "Mean"])

mu_hat <- beta1_hat + beta2_hat * (transistor$Year - transistor$AvgYear)
std_resid <- abs((transistor$logC - mu_hat) / sigma_hat)

transistor[which.max(std_resid), "Processor"]
```

```
[1] "F21"
```